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3.4: Research Publications and Awards

3.4.3: Number of research papers published per teacher in the Journals as notified on UGC CARE list during the last five years

Research Journal Paper published in Year Jan-Dec 2019

Sr. No	Research Paper Title	Author Names	Link to Paper
1	An optimized facial stimuli paradigm for hybrid SSVEP+P300 brain computer interface	Dhananjay Kalbande, Deepak Kapgate, Urmila Sharwankar	https://europepmc.org/article/med/31298507
2	Adaptive classification to reduce non-stationarity in visual evoked potential brain-computer interfaces, Bio-Algorithms and Med-Systems	Dhananjay Kalbande, Deepak Kapgate, Urmila Sharwankar	https://www.degruyter.com/document/doi/10.1515/bams-2019-0020/html
3	Cost Optimized Hybrid System in Digital Advertising Using Machine Learning	Dhananjay Kalbande, Surekha Dholay, Swati Kulkarni and Avinash Sharma	https://www.ijitee.org/wp-content/uploads/papers/v8i8s2/H11580688S219.pdf
4	Advanced credit score calculation using social media and machine learning	Sudhir N. Dhage, Kulkarni Swati	https://content.iospress.com/articles/journal-of-intelligent-and-fuzzy-systems/ifs169948
5	Survey on Virtual Assistant: Google Assistant, Siri, Cortana, Alexa	Dhage S.N, Tulshan A.S.	https://www.researchgate.net/publication/330165159_Survey_on_Virtual_Assistant_Google_Assistant_Siri_Cortana_Alexa_4th_International_Symposium_SIRS_2018_Bangalore_India_September_19-

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6	A Survey on Word Embedding Techniques and Semantic Similarity for Paraphrase Identification	Anant V. Nimkar, Divesh R. Kubal	https://doi.org/10.1504/IJCSYSE.2019.098417
7	Cost Optimized Hybrid System in Digital Advertising Using Machine Learning	Surekha Dholay, Swati Kulkarni, Dhananjay Kalbande, and Avinash Sharma	https://www.ijitee.org/wp-content/uploads/papers/v8i8s2/H11580688S219.pdf
8	SOI FinFET Based Instrumentation Amplifier for Biomedical Applications	Reena Sonkusare, Omkar Joshi and S. S. Rathod	https://www.sciencedirect.com/science/article/abs/pii/S0026269219302277
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10	SOI FinFET based instrumentation amplifier for biomedical applications	Reena Sonkusare, Omkar Joshi, S.S. Rathod	URL: https://in.booksc.eu/book/76173506/7cd888
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12	Musical Instrument Tuner Asian Journal of Convergence in Technology	Upasana Thakuria, Shreeram Narayanan, Reena Kumbhare	https://asianssr.org/index.php/ajct/article/download/875/693
13	Vehicle Number Plate Detection using Convolutional Neural Networks	Yash Patil, Krishna Mehta, Sopan Phaltankar, Dr. Reena Kumbhare	https://asianssr.org/index.php/ajct/article/view/868
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An optimized facial stimuli paradigm for hybrid SSVEP+P300 brain computer interface.

Kapgate D ¹✉, Kalbande D ², Shrawankar U ³

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Abstract

Background

In this paper, hybrid brain-computer interface (BCI) that merges event related potentials (P300) and steady state visual evoked potentials (SSVEP) simultaneously is proposed to enhance the performance of four-choice computer system. Here we integrate human facial structure in external visual stimulus to evoke stronger cortical response in hybrid SSVEP+P300 BCI and compare it with non-face stimuli. We also addressed question of feasibility of eliciting one potential by face and other by non-face stimuli. To evoke SSVEP and P300 responses, paradigms with non-face stimuli, neutral face stimuli, and facial expression changes stimuli are proposed. We also projected additional paradigm where SSVEP is elicited by non-face and P300 by flashing different facial expressions. Results proved that last paradigm evokes stronger cortical potentials and thereby improves system accuracy and ITR than other paradigms. We discuss external stimulus parameters that might affect simultaneous evocation of multiple brain potentials and their deterioration effect on individual potentials.

Methods

Ten healthy volunteers (seven male, aged 22-28 years, mean 25.6 years) contribute in our experiment. LCD screen with refresh rate of 60 Hz is used for external stimulus and all trials are performed in darkened room.

Results

The ideal BCI system should have maximum ITR and comfortable rating. We have proven among our paradigms that emotional and flickering paradigm has maximum comfortableness with highest ITR.

Conclusions

In this paper, we have proposed four paradigms as non-face paradigm, face neutral paradigm, face emotional paradigm, and face-emotional and flicker paradigm. We have tested all paradigms and compared them with each other about potential response strength, accuracy, ITR and comfortableness. Results show that face-emotional flicker paradigm performed better in all aspects than all four paradigms.

Full text links

Citations & impact

Adaptive classification to reduce non-stationarity in visual evoked potential brain-computer interfaces

Deepak Kapgate , Dhananjay Kalbande and Urmila Shrawankar

From the journal Bio-Algorithms and Med-Systems

<https://doi.org/10.1515/bams-2019-0020>

Citations

1

Abstract

Non-stationarity of electroencephalogram (EEG) signals greatly affect classifier performance in brain-computer interface (BCI). To overcome this problem we propose an adaptive classifier model known as extended multi-class pooled mean linear discriminant analysis (EMPMLDA). Here, we update the average class pair co-variance matrix along with pooled mean values. Evaluation of classifiers are done on visual evoked cortical potential data-sets. We demonstrate that EMPMLDA can significantly outperform other static classifiers such as MLDA and adaptive classifiers (MPMLDA). Furthermore an optimal update coefficient can be achieved using different datasets.

Keywords: adaptive classification; brain-computer interface; non-stationarity; visual evoked potentials

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Ethical Approval: The conducted research is not related to either human or animal use.

Author contributions: All the authors have accepted responsibility for the entire content of this submitted manuscript and approved submission.

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Cost Optimized Hybrid System in Digital Advertising using Machine Learning

Avinash Sharma, Swati V. Kulkarni, Dhanajay Kalbande, Surekha Dholay

Abstract: Digital advertising or Internet marketing is the term used to describe the process of advertising a product or brand through digital medium. It includes promotional advertisements and messages delivered through email, social media websites, search engines, mobile applications, web sites and affiliates programs. This work presents a system that solves the challenge of reaching correct people and optimizing the cost problem using various machine learning techniques. It also explains various research trends in predictive analytics, product pricing and targeting audience for digital advertising. Digital Advertising has captured wide attention from market. It is very powerful tool to reach correct people at correct time. Also it reduces the cost of broadcasting advertisement as the ad is displayed only to people who might be interested in the content. The interest prediction for audience targeting provides 89.44% accuracy using Naive Bayes classifier.

Index Terms: Digital Advertising, Predictive Analytics, Cost Optimizing, Audience Expansion, Machine Learning, Logistic Regression, Naive Bayes.

I. INTRODUCTION

Advertising industry has seen tremendous transformation in digital era. It started from displaying an advertisement on the wall of church. It has now reached to personalize advertisements on individual's social networks. Approximately 20 years ago an advertisement used to mean displaying, promoting a product or service through broadcasting information about the same through newspapers or television. This was then the only way to attract people towards the product or service and gain probable customers. As the technology advanced, the old broadcasting way of advertisement changed to personalize digital advertisements.

Nowadays advertisers promote their product using online social platforms such as Facebook, YouTube and many other websites. Machine Learning and Artificial intelligence is now widely used to understand the current trends and to know the characteristics of people who might be interested in the contents of their advertisement. Machine learning algorithms can be used in many ways to achieve impressive personalize experience. The three most popular ways are predictive analytics, cost optimizing and audience targeting. The use of machine learning techniques in digital advertising is beneficial for all three stakeholders namely advertiser, publisher and end users. It is very effective and cost saving option because it gets more recognition with lesser number of displays.

Predictive analytics is the process of understanding one's future behaviour depending on the past actions. In predictive analytics, machine learning algorithms can be applied to predict CTR (Click-Through-Rate: No. of Clicks per ad impressions) for an advertisement. CTR value is not only used for targeting specific audience and cost cutting but also depicts brand reputation and ranking in global competition. Hence it is now very important for companies to achieve high CTR for their advertisement. This is the reason industry and academia showing deep interest in research for digital advertising domain.

Cost optimizing is a technique for optimizing pricing for sales and profit. Profit Optimizing is the process of gaining maximum profit through minimum advertisement displays. Cost for displaying an advertisement depends on various factors such as cost per click, inventory that is space provided by publishers for advertisement display, CTR etc. This can be achieved by real time process of estimating utility (like clicks, conversions) of ad impressions, forecasting the market value of ad impression hence the cost of product. Profit Optimization can be achieved through machine learning in predictive analytics and audience expansion.

Audience Targeting Techniques is the process of predicting relevance of advertisement to individual. Machine learning algorithms perform analysis on individual's personal data that they voluntarily share with companies. The results are promising for audience targeting. The research in this area is done by the team working at LinkedIn Corporation. They deployed their audience expansion model on LinkedIn's social networking platform. This large scale experiment demonstrated effective results that helped not only advertiser, and LinkedIn as social networking platform but also end users. Companies like Google, Facebook also does audience targeting for

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Sudhir N. Dhage

Advanced credit score calculation using social media and machine learning

Authors Swati V Kulkarni, Sudhir N Dhage

Publication date 2019/1/1

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Description The Credit Score is the most fascinating three digit number associated with an individual or an organization as it figures out what loans you will qualify for and the interest rate you will pay. Current credit scoring system is based on the financial history of individual or organization. This work illustrates a new credit scoring system which incorporates Legacy credit score and emotional/social credit score. The legacy credit score is based on the financial history of an individual. The emotional/social credit score is based on analysis and study of social media and other web interaction. The new system called information trustworthiness is developed to improve the precision of social media data when compared with data from reliable sources. Finally, the proper fractions of legacy credit score and emotional/social credit score are added to get Advanced Credit Score. This score is more precise than Legacy credit score as it ...

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Scholar articles Advanced credit score calculation using social media and machine learning
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Abstract. Virtual assistant is boon for everyone in this new era of 21st century. It has paved way for a new technology where we can ask questions to machine and can interact with IVAs as people do with humans. This new technology attracted almost whole world in many ways like smart phones, laptops, computers etc. Some of the significant VPs are like Siri, Google Assistant, Cortana, and Alexa. Voice recognition, contextual understanding and human interaction are the issues which are not solved yet in this IVAs. So, to solve those issues 100 users participated a survey for this research and shared their experiences. All users' task was to ask questions from the survey to all personal assistants and from their experiences this research paper came up with the actual results. According to that results many services were covered by these assistants but still there are some improvements required in voice recognition, contextual understanding and hand free interaction. After addressing these improvements in IVAs will definitely increased its use is the main goal for this research paper.

Keywords: Human factors · Virtual Personal Assistant · Smart devices

1 Introduction

Intelligent Virtual Assistant (IVA) is "an application that utilization information, for example, the user's voice... and logical data to give help by noting inquiries in normal dialect, making suggestions and performing activities". Inside the writing the term IVA is utilized conversely with terms, for example, Conversational Agents, Virtual Personal Assistants, Personal Digital Assistants, Voice-Enabled Assistants or Voice Activated Personal Assistants, to give a few illustrations. IVAs join talk affirmation lingo understanding, trade organization, tongue age and talk association to respond to clients request and sales. Voice engaged IVAs like Siri, Google Assistant, Microsoft Cortana and Amazon Alexa are by and large open on cutting edge cell phones, and continuously in homes (e.g. Amazon Echo and Google Home) and automobiles (e.g. Google Assistant blend with Hyundai). The market for IPAs is foreseen to reach 4.61 billion by the mid 2020s.

The specialized foundations that empower IVAs have progressed quickly lately and have been the subject of broad research. Be that as it may, investigate concentrated on understanding the users experience of IVAs is more restricted. Obviously, from our point of view, this has had suggestions for the selection and utilization of IVAs. For

 NO ACCESS

A survey on word embedding techniques and semantic similarity for paraphrase identification

Divesh R. KubaI and Anant V. Nimkar

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ABOUT

Abstract

In natural language processing (NLP), paraphrase identification (PI) determines the relatedness between the pair of sentences having fewer or negligible lexical overlap but still pointing towards the same meaning. The major challenge faced while attempting to solve this problem is the many possible linguistic variations conveying the same purpose. This paper aims to provide a detailed survey of traditional similarity measures, statistical machine translation metrics, machine learning and deep learning techniques and a well-defined flow between them. This article encompasses various word embedding methods and step-wise derivation of its learning module. This survey paper also provides a definite flow pointing towards the evolution of deep learning in an unambiguous manner. A comparative analysis of various techniques to solve PI is presented and it will provide research directions to work in the similar domain.

Keywords

paraphrase identification, word embedding, deep learning, convolutional neural network, CNN, semantic similarity

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SOI FinFET based instrumentation amplifier for biomedical applications

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Keywords:

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DC gain
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Strong and weak inversion region

ABSTRACT

In this paper, we present a design of SOI FinFET device based Instrumentation Amplifier (IN-Amp) for biomedical applications. An IN-Amp is operated in weak inversion regime effectively for low power and low voltage applications. The performance metrics of the proposed IN-Amp such as DC gain, Common mode rejection ratio (CMRR) were enhanced by using nanometre FinFET device technology. To achieve these desirable performance metrics, size of transistors is optimized and passive resistances are replaced by active 30 nm SOI FinFET transistor. A FinFET based DC offset removal circuit is designed to reduce the offset voltage at the output of the IN-Amp. The output achieved on appending DC offset removal circuit is exact replica of input over wide range of frequency, which can be adjusted by varying clock inputs to the circuit. It is observed that the proposed SOI FinFET based Instrumentation amplifier achieves an overall gain and CMRR of 99.6 dB and 103 dB respectively.

1. Introduction

An Instrumentation amplifier (IN-Amp) is widely used in industrial and medical applications, where extremely low power and low voltage signal needs to be conditioned in the presence of large common-mode voltages and DC potentials. It is frequently used in biomedical applications involving measurement of low frequency, Microvolt signals like ECG, nerve synapses pulses, brain signals etc. IN-Amps acquire these microvolts signals from electrodes in the presence of numerous physiological and environmental interferences. These amplified signals are processed using nanometre technology in an extremely low power mode. Current focus in VLSI design uphold the continued validity of Moore's law by continuous miniaturization in deep sub-micron technologies [1]. Analog, digital and RF blocks can be integrated on a single chip substrate using System on Chip (SoC) technology. Sub-micron nodes present unique challenges in analog circuit design [2]. The analog performance of the devices significantly influence the Figure of Merit of analog circuits. At shorter channel lengths, effects of parasitic and transistor mismatch become significant [3]. The planar gate size of MOSFET is reduced substantially at smaller gate lengths affecting the device transconductance (g_m) and intrinsic gain. To improve the performance parameter of analog circuits like DC gain (A_{DM}), Common mode gain (A_{CM}), Unity Gain-Bandwidth and Common Mode Rejection Ratio, we need to optimize the geometrical dimensions of device in nanometre regime.

Fundamentally, the input signals of any mixed mode processing system is analog in nature [4]. There are multiple challenges one can face while integrating the devices in the analog circuits at lower technology nodes, such as standard CMOS processes cannot be scaled beyond 50 nm due to Short Channel Effects (SCE) and Drain Induced Barrier Lowering (DIBL) [5,6]. Hence to overcome these limitations traditional architecture should be replaced by new technology. Silicon on insulator (SOI) fin shaped field effect transistor (FinFET) structure [Fig. 1] has been in the last few years the object of intensive research. Moreover, most of the research work reported in analog domain is based on nanometre CMOS technologies [3]. SOI technology provides a low capacitance, low voltages which enables high-speed operation and reduces the power consumption. Moreover, the focus of this paper is to design low power, high CMRR and low noise IN-Amp using nanoscale technology, such as FinFETs.

In this paper, we simulate and investigate the performance of 30 nm SOI FinFET with high- κ gate dielectric based instrumentation amplifier for amplifying very weak biomedical signals. An IN-Amp circuit consists of input buffering stage followed by a high gain differential stage, hence can be used further in analog domain. It is designed in strong inversion as well as weak inversion region. The weak inversion region operation of FinFET device provides very high transconductance as compared to CMOS, which can be utilized to boost gain for weak biomedical signals. The weak inversion region of operation enables low power consumption as it sets speed limitations for device operation. However, biomedical

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Bandwidth Distribution Algorithm of Applications in LTE-Femtocell Network

Anita Jadhav, Y. Srinivasa Rao

Abstract: The wireless communication has become the essential of today's Internet of Things (IoT) and mobile applications which is increasing exponentially in near future. These applications demand high data rate to speed up the communication among IoT or internet applications. The data rate requirement of individual application is different. To fulfill these demands of every application Long Term Evolution (LTE) integrated with femtocell technology is identified as suitable wireless technology. The running applications on a user equipment changes dynamically. Hence in dense deployment of femtocell and IoT devices, the bandwidth distribution to many active applications becomes more complicated. For multiple running applications per device with available bandwidth constraint, optimal bandwidth allocation to applications has become the most challenging issue. We include the bandwidth allocation to applications during handover of user equipment in LTE-femtocell network. The handover procedure needs efficient cell selection criteria for target cell selection.

Hence in our research we focus on bandwidth distribution among applications by proposing "Efficient Bandwidth Distribution" (EBD) mechanism and efficient cell selection by "Dynamic Cell Selection" (DCS) mechanisms to mitigate above mentioned issues.

Keywords: LTE, femtocell, EBD, DCS, Throughput, QoS, 3GPP.

I. INTRODUCTION

The capacity of wireless networks must grow continuously in order to satisfy the demand of traffic from users and applications. To support these requirements of wireless network, Long-Term Evolution (LTE) and LTE-Advanced (LTE-A) have been standardized by 3GPP [1]. LTE is targeting data rate of 100Mbps/s and 50Mbps/s in the downlink and uplink respectively. The improvements in data rate are due to enhanced channel-dependent scheduling and rate adaptation. The data rate also improved in frequency domain by spatial multiplexing with MIMO, and larger channel bandwidths of up to 20MHz [6]. The standardization of new physical air interfaces is targeting downlink peak data rates of 1Gbit/s at low mobility and 100Mbps/s at high mobility.

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Combining the principles of Network Functions Virtualization (NFV) with Software Defined Network (SDN) benefits for programmability and flexibility to the fore [7]. The offloading of software defined radio processing to general processing is achieved through NFV to meet the flexibility of programming [5]. Special emphasis to the 4G cellular Long-Term Evolution network is given through the concept of Self-Organizing Network (SON). It is observed that huge amount of data is generated in the 4G networks and expected more in future due to densification. The future 4G or 5G network management is expected to provide a new set of challenges due to the need to manage very dynamic networks, need to support 1000x traffic, and 10 fold users, improve energy efficiency, need to improve the experience of the users by enabling Gbps speeds and reduced latency, need to handle heterogeneous spectrum access privileges through the LTE. In our paper we will only focus on the applications to 4G network management to meet high data rate, bandwidth optimization, handover management especially during handover and to reduce delay, bit error rate in the LTE network integrated with femtocell. In our work in order to gain higher data rate we optimize the femtocell-femtocell handover procedure in a macrocell area. To optimize the handover using bandwidth and power parameter, we proposed Dynamic Cell Selection (DCS) mechanism. Our DCS helps in load balancing among the neighboring cells by evaluating the measurement reports and selecting the suitable cell which has sufficient resources. The bandwidth measurement in DCS is computed by bandwidth requirement of users and their applications. To calculate the users and applications bandwidth requirement we proposed Efficient Bandwidth Distribution (EBD) mechanism. Our innovative models, the DCS and EBD mechanisms are simulated in NS-3. We studied and evaluated the performance of our models in terms of throughput, bandwidth, user satisfaction rate for LTE integrated with femtocell Networks. We extended our work by another approach that is MIMO antenna technique to achieve higher data rate and larger channel bandwidths of up to 20MHz. The MIMO technique is used in real time implementation for end-to-end communication using Software Define Radio. The AD9361 provides flexible sampling rate which is very wide ranging from 547 Ksps to 61.44 Msps [18]. The end-to-end communication is shown for image transmission over-the-air using MIMO 2x2 and 1x1 antenna configuration with 64 QAM Modulation schemes and 10 MHz bandwidth of LTE. Due to the limited hardware devices, we show the cell search mechanism and end-to-end communication between two user equipment.





SOI FinFET based instrumentation amplifier for biomedical applications

Reena Sonkusare^{a,*}, Omkar Joshi^b, S.S. Rathod^a

^a Sardar Patel Institute of Technology, Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai 400058, India

^b Arizona State University, Tempe Campus, Tempe AZ-85281, United States

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Keywords:

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ABSTRACT

In this paper, we present a design of SOI FinFET device based Instrumentation Amplifier (IN-Amp) for biomedical applications. An IN-Amp is operated in weak inversion regime effectively for low power and low voltage applications. The performance metrics of the proposed IN-Amp such as DC gain, Common mode rejection ratio (CMRR) were enhanced by using nanometre FinFET device technology. To achieve these desirable performance metrics, size of transistors is optimized and passive resistances are replaced by active 30 nm SOI FinFET transistor. A FinFET based DC offset removal circuit is designed to reduce the offset voltage at the output of the IN-Amp. The output achieved on appending DC offset removal circuit is exact replica of input over wide range of frequency, which can be adjusted by varying clock inputs to the circuit. It is observed that the proposed SOI FinFET based Instrumentation amplifier achieves an overall gain and CMRR of 99.6 dB and 103 dB respectively.

1. Introduction

An Instrumentation amplifier (IN-Amp) is widely used in industrial and medical applications, where extremely low power and low voltage signal needs to be conditioned in the presence of large common-mode voltages and DC potentials. It is frequently used in biomedical applications involving measurement of low frequency, Microvolt signals like ECG, nerve synapses pulses, brain signals etc. IN-Amps acquire these microvolts signals from electrodes in the presence of numerous physiological and environmental interferences. These amplified signals are processed using nanometre technology in an extremely low power mode. Current focus in VLSI design uphold the continued validity of Moore's law by continuous miniaturization in deep sub-micron technologies [1]. Analog, digital and RF blocks can be integrated on a single chip substrate using System on Chip (SoC) technology. Sub-micron nodes present unique challenges in analog circuit design [2]. The analog performance of the devices significantly influence the Figure of Merit of analog circuits. At shorter channel lengths, effects of parasitic and transistor mismatch become significant [3]. The planar gate size of MOSFET is reduced substantially at smaller gate lengths affecting the device transconductance (g_m) and intrinsic gain. To improve the performance parameter of analog circuits like DC gain (A_{DM}), Common mode gain (A_{CM}), Unity Gain-Bandwidth and Common Mode Rejection Ratio, we need to optimize the geometrical dimensions of device in nanometre regime.

Fundamentally, the input signals of any mixed mode processing system is analog in nature [4]. There are multiple challenges one can face while integrating the devices in the analog circuits at lower technology nodes, such as standard CMOS processes cannot be scaled beyond 50 nm due to Short Channel Effects (SCE) and Drain Induced Barrier Lowering (DIBL) [5,6]. Hence to overcome these limitations traditional architecture should be replaced by new technology. Silicon on insulator (SOI) fin shaped field effect transistor (FinFET) structure [Fig. 1] has been in the last few years the object of intensive research. Moreover, most of the research work reported in analog domain is based on nanometre CMOS technologies [3]. SOI technology provides a low capacitance, low voltages which enables high-speed operation and reduces the power consumption. Moreover, the focus of this paper is to design low power, high CMRR and low noise IN-Amp using nanoscale technology, such as FinFETs.

In this paper, we simulate and investigate the performance of 30 nm SOI FinFET with high- κ gate dielectric based instrumentation amplifier for amplifying very weak biomedical signals. An IN-Amp circuit consists of input buffering stage followed by a high gain differential stage, hence can be used further in analog domain. It is designed in strong inversion as well as weak inversion region. The weak inversion region operation of FinFET device provides very high transconductance as compared to CMOS, which can be utilized to boost gain for weak biomedical signals. The weak inversion region of operation enables low power consumption as it sets speed limitations for device operation. However, biomedical

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COMPARISON OF FILTER OUTPUTS FOR ECG SIGNALS

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Abstract

ECG (Electrocardiogram) is the graph of voltage recorded from the electrical impulses generated by the heart versus time. It typically consists of the P-wave, the QRS complex and the T-wave. They are used to determine the healthiness of a patient's heart. In a medical environment, there is a need to filter out the noise added throughout this signal to generate a precise rendition of a patient's ECG. This has been attempted through the use of FIR high-pass, low-pass and band-pass filters. Their output has been compared on the basis of parameters like Signal-to-noise-ratio (SNR). The output through the filters was compared with the noisy input by the method of correlation to determine how much noise had been attenuated. Thus, the optimal filter was decided.

Keywords : Electrocardiogram, filter, low-pass, high-pass, band-pass

phaseresponse. The types of FIR filters applied here are the lowpass, highpass and bandpass filters. In low-pass filters, there is a threshold frequency specified called the stopband frequency above which all the signal is attenuated. The high-pass filter has a threshold frequency also called the stopband frequency below which all the signal is attenuated. The band-pass filter has two passband frequencies. The signal below and above this range of frequencies is attenuated.

INTRODUCTION

As stated above, an ECG denoted the electrical activity of a human heart. It is a periodic wave with typical characteristics that can help differentiate between a healthy and a diseased heart. The range of frequency of an ECG signal is usually from 0.05 Hertz to 100 Hertz with the amplitude ranging from 1 mV to 10 mV.

The ECG is measured with the help of ten electrodes attached across the body to measure the polarisation and depolarisation with respect to blood flow. The outlined process to measure the ECG signal can introduce two types of noise[1]. The first type is noise with a high frequency and consists of white Gaussian noise and Electrocardiogram and the second type is low frequency noise. Our experiment aims at reducing additive white Gaussian noise. With the aid of FIR filters [1] of the type low-pass, high-pass and band-pass with fixed passband and stopband frequencies, the ECG signal with noise is brought back from degradation and the outputs are compared [2].

RELATED THEORY

The results of the experimentation depend on the working of the FIR filters used. Finite Impulse Response filters accept delayed input signal to give an almost linear

PROPOSED SYSTEM DESIGN

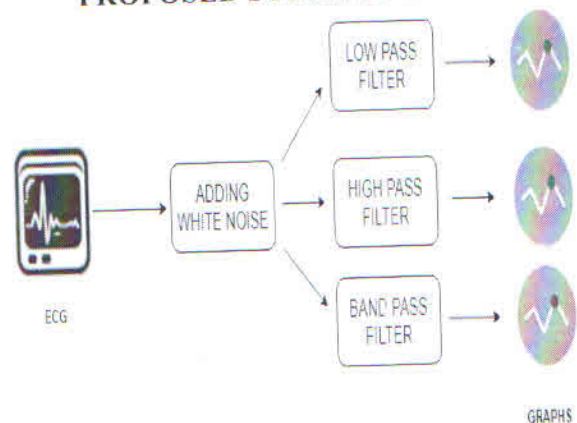


Fig. 1. Block Diagram.

MATLAB

The software used to run the simulation for the specified setup was MATLAB (matrix laboratory). MATLAB [3] provides a convenient and easy-to-use interface and environment to run the simulation and view the desired output for our analysis.

ECG Signal

The ECG [4] signal is distinctively characterized by P, Q, R, S, T and U points which make up the P wave, the QRS complex, the T wave and the rarely seen U wave. Each wave is a result of the polarisation, depolarisation and repolarisation of the heart's atria and ventricles.

Musical Instrument Tuner

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Abstract—There are numerous types of signal processing with audio signal processing being a major sub-set. As the name suggests, audio signal processing is the processing of audio signals which are sound waves in their electronic form. This has wide applications like analog-digital format conversion, removal of background disturbance (noise) for clearer audio, addition of required effects, and many others. According to a number of studies, there is an acute need of representing musical audio signals in the form of their fundamental frequencies for the purpose of tuning. This paper proposes a moderate accuracy level representation of an audio file on the basis of power spectral density plot in order to determine whether the given note is flat or sharp that is to determine and verify a chord via audio signal processing on the basis of power or frequency spectral density plot using MATLAB. Result analysis of the fundamental frequencies was completed and compared with the values of fundamental frequencies of various musical instrument notes from the database chart. In this paper, we have used the Welch Method and the Periodogram Method to compute the power spectral density.

Index terms—Audio signal processing, Welch power spectrum, Periodogram power spectrum, fundamental frequency, audio octave level, audio string.

I. INTRODUCTION

The human auditory system, with a hearing range of 20Hz-20kHz, is capable of processing complex mixture of sound and identify the sources and environmental abstractions. Such a process is called auditory scene analysis. But it is not sensible enough to always determine correctly with accuracy if for given audio octave level-string played is sharp or flat. This is where the problem arises for the musicians. People operating with musical instruments have a problem in determining whether the tune or note that they are playing is flat or sharp. For a given audio octave level, if the frequency of the note goes higher than its required fundamental frequency, it is called a sharp note. Similarly, if it goes lower than its required fundamental frequency, it is called a flat note. It makes a significant difference if flat note of a string is played instead of its sharp note and vice versa i.e. G flat instead of G sharp results in lower frequency note which sounds out of tune while playing a musical instrument. There are a number of power spectral density types to estimate the fundamental frequencies of a given audio file- Welch method, periodogram method, covariance method, Burg method among others.

II. THEORY

A. Welch's Method

Welch's Method is used for estimating the power of a given signal at different frequencies. The spectrum is a result of converting a signal from time domain to frequency domain. The noise generated is reduced in the Welch's method of spectral density estimation. [1]

B. Fast Fourier Transform

A signal consists N number of complex points in the time frequency system, these points contains two values that is real and imaginary. The N point time domain signal is obtained by decomposing the N time domain signal each of them have single point. The next step is of finding the N frequency spectra of the obtained N time domain signal and the end step is synthesizing the the N spectra into a single frequency spectrum, this concept can be understood by a simple example of 16 point signal which is decomposed in 4 different stages, here we first make 16 point signal into two, 8 point signal then in next step these two 8 point signal is decomposed into 4 signals of 4 point. This process keeps going on till there are N signal having single point. The decomposition used for cutting the signal in two is called as interlaced decomposition.

III. METHODOLOGY

Audio signal processing [2] of the played note was carried out on MATLAB software. The input is an audio file which will be obtained by recording the voice note played by the string instrument. The audio signal can be in any any format such as .mp3, .wav, .wma, .ogg among others. The audio signal was converted into audio samples for further processing. The Welch Power Spectral Density Analysis helps in determining the fundamental frequency of the note. The spectrum was analysed and observed. The fundamental frequency obtained by the spectral analysis of the audio signal was then compared with the chord frequency chart shown in Fig. 1. If the obtained frequency of the voice note is greater than the fundamental frequency of the chord then the instrument is sharp, similarly if the obtained frequency is lesser than the fundamental frequency then it is flat. Thus, we computed whether the instrument is in tune or out of tune. The Fast Fourier Transform of the voice signal was also observed and analysed [3]. The code for determination of fundamental frequency was written on MATLAB software and the required

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Vehicle Number Plate Detection using Concolutional Neural Networks

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Abstract—Vehicle Number Plate Detection (VNPD) Security is one of the main concerns in today's world. Every residential and government premises takes logs of the cars which enter their compounds. The manual logging of visiting vehicles in residential, commercial, government premises is cumbersome and expensive. An automated process for the same will greatly increase efficiency and reduce costs. This paper describes a seamless method for the same. The developed system will first detect the vehicle and then captures the vehicle image. Vehicle number plate region is extracted using 2D convolution. From the number plate obtained from the convolution the image frame is masked into grey scale. After masking, the trained model detects the texts in the frame and prints it on the screen.

Keywords—Image Processing, Convolution, Convolutional Neural Network, TensorFlow, OpenCV, Python

I. INTRODUCTION

The global spending on residential security solutions reached a staggering 91.4 billion dollars in the year 2018 which is a 10 percent increase from the year before. The crime rate in the world is ever increasing and the residential complexes in the cities are making sure every car which goes in or out of the complex is logged in. Watchmen are the ones writing down the number of each car which enters the premises. The average salary per watchman in an Indian city is around 8000 rupees. This system is not very efficient, and it also forces the car to stop at the gate which also makes it time consuming

Most of the Government, residential as well as corporate complexes have 24-hour CCTV surveillance to record any theft which might take place. Setting up such an extensive system with its own memory is expensive and cumbersome. On top of this system there is also a need to keep people to log in the vehicles which adds on to the cost. If the logging system was to be made automated it would greatly reduce the cost of security for any building complex. The video feed from the CCTV cameras which are already installed at the gates can be used to make this VNPD (Vehicular number

plate detection) system which automatically logs in the number plate of the car which enters or exits the gate.

II. OBJECTIVES

The main objectives of VNPD are:

A. Automate the entry system of vehicles

The entry of vehicles is presently done manually. VNPD aims to automate this manual task.

B. Reduce the security expenses

This system will reduce the cost to hire people to write down the number plates of the cars entering the complex.

C. Increase in speed of logging

The vehicles entering the gates will not have to stop which will save time and increase of the vehicles entering.

D. Increase efficiency

The manual logging of number plates can have human errors. The automation of the system will eliminate this problem.

E. No need for physical storage

The data can be stored on the cloud and the need to keep physical books won't remain anymore. This is more reliable as the chances of the data getting lost or corrupted is low compared to a physical book.

III. SYSTEM DESIGN

A. Python

Python is a high-level object-oriented, programming language with dynamic semantics. The reason behind using python for VNPD is its support for extensive number of libraries helpful in machine learning, deep learning and data preprocessing [1].

B. OpenCV

OpenCV is an open source computer vision library. It provides many functionalities which can be used to work

An Approach for Predication of Complications Women Fights During Maternal Period

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Date Written: 2019

Abstract

Pregnancy and motherhood are natural processes in the lives of women. One of the great feelings for every woman is begin a Mother, but maternal period is very much susceptible for health issues. Some old illnesses may increases or new illnesses can start during this period. Recently Indian health status has improved. Child mortality rates and maternal mortality are declined in current scenario but when Indian situation is compared with other developed countries it is not so good. Advance in data analytics has become one of the driving forces in multiple fields to turn them into technology dependent ones. One of the fields where this impact is still at lower level is maternal health care. There is a need to analyze the symptoms, take steps towards the precautions and dealing with the complication which women fights during this period. This paper discuss various

Anticipating Pointless ER Visits to Lessen Human Services Costs

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Abstract: The sole reason for the Emergency Room (ER) is to spare lives by furnishing quick consideration regarding individuals with hazardous circumstances. With 24×7 access, a wide exhibit of administrations, and the most recent innovation close by, ER groups are very much prepared and prepared in treating therapeutic urgencies, balancing out patient conditions, and counteracting further harms.

Keywords: ER, ED, Minnesota, NYU, Billings/Ballard

I. INTRODUCTION

An expanding maltreatment of ERs, either because of patient numbness or comfort, requests pressing consideration from the two payers and suppliers. There is a great deal of archived proof where patients have utilized ERs for circumstances that could have been treated in a more financially savvy care setting, for example, Urgent Care Clinics (UCCs) or Patient-Centered Medical Homes (PCMHs).

II. REASON FOR OVERCROWDED ER'S

An increasing volume of high-acuity patients presenting to the ED insufficient inpatient capacity. [1] A. Insights of reduces unnecessary ER visits

This paper shows a far reaching start to finish answer to lessen ER usage for non-rising conditions. The proposed information driven arrangement use prescient investigation to build up a system to distinguish individuals prone to utilize the ER for avoidable reasons sooner rather than later, and the arrangement prescribes structuring explicit intercessions to counteract future visits. Amid the procedure, we will have likewise assembled a case to use examination in a spry method to quickly infer greatest esteem.

A. Maintaining the Integrity of the Specifications

Having medical coverage is imperative for a few reasons. Uninsured individuals get less therapeutic consideration and less convenient consideration, they have more regrettable wellbeing results, and absence of protection is a financial weight for them and their families. Additionally, the advantages of growing inclusion exceed the expenses for included administrations. Wellbeing net consideration from medical clinics and facilities improves access to mind yet does not completely substitute for health care coverage. These discoveries are upheld by much research, albeit a few alerts are fitting in utilizing these outcomes. [2]

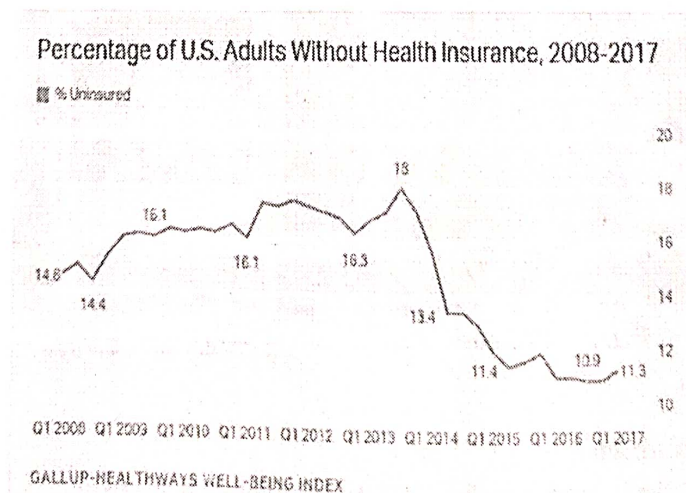


Fig. 1: Percentage of uninsured people in USA (2008-2017)[3]

Sentiment Analysis of Mobile Product Reviews

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Abstract: Reviews about products have started playing an important role in finding opinions about products that are available in the market [1]. Those reviews spread like rumours and help in decision making for the consumers to purchase the product or not and as well to the manufacturers to get feedback about their products. The reviews help in better selection of products for the customer and hence help them in selecting the best and the appropriate phone for their use. Several researches have been conducted on this problem using various techniques and have produced results of varied kinds. Many different kinds of approaches have been used to conduct the study of this problem and have shown the various desired results about a particular smartphone product. Some researches show comparative studies about various methodologies used for solving the problem while some make a combination of various approaches into one solution. All of these have been successful in their own ways and have helped the user in determining the best product according to their requirements. The growing number of opinions about a particular product make it ambiguous for a customer to take a decision of whether to buy that product or not [1]. The problem customers face while picking a mobile phone is that there is no mechanism which gives detailed reviews about a product. Taking the previous works into account we have tried to enhance the research about this topic by adding some contribution in our own way and helping the community. We make an effort here to build a system that efficiently tries to predict the sentiments regarding a mobile product based on the reviews that people have given to that product on a particular platform.

I. INTRODUCTION

Reviews provided by users prove to be vital for customers interested in buying a particular product. Reviews provided by customers already using a particular product come with a much broader sense of classification in terms of functionality and user experience. One more advantage that comes along with the product reviews is that it helps the product manufacturers to analyze response their product is getting in the market [2]. Its not possible for any user to read every review and make a decision [3]. The solution we figured out for this problem is to make a summary of all the reviews and produce a comparative analysis of all the reviews. This will not only help the customers get a comparative study about the mobile products features but also, they will also get to know about the user experience and the pros and cons of the product they are looking to buy.

The way in which this is achieved is that first the reviews from multiple e-commerce platforms are been extracted and then the information is cleaned [4]. The cleaned data is then passed through a test where they are classified based on various categories like positive or negative, hit or flop etc. and depending on the classification the conclusion is derived whether the product is a success or a failure [4]. This can be done individually for every review or together for all the reviews as one. The emphasis is to capture different aspects of a mobile product in order to make a broader analysis of the product.

Sentiment analysis can be termed as the application of text analysis, natural language processing, biometrics and computational linguistics to comprehensively associate, draw, quantify, and examine affective states and instinctive information [5]. Sentiment analysis is widely recognized to voice of the customer medium such as reviews and survey reactions, online and social network media, and healthcontrol materials for applications that range from customer service to marketing to clinical medicine [5]. The three main categories to into which sentiment analysis can be classified is: viz. statistical methods, knowledge-based techniques and hybrid approaches[6]. Sentiment Analysis is circumstantial extraction of text which pin-points and pulls out instinctive information in origin material, and helping a business to perceive the social sentiment of their product, brand or service while tracking online conversations [7].

It is also the most general text classification tool that examines an incoming message and tells about the positivity, negativity or neutrality of the underlying sentiment. It helps in social network media exploration and allows us to gain an outline of the wider public belief of certain topics. Sentiment analysis on large collections of data set including web pages, internet discussion groups, online news, online reviews, social media and web blogs is automatically possible due to huge availability of free or paid open source sentiment analysis tools that deploy machine learning, statistics and natural language processing techniques. [8]. The most extensively used approach to achieve this is to distinguish the data with two subsets of data one of which contains positive words and the other most of the negative words.

Text summarization refers to the expertise of condensing extensive chunks of scripts. The objective is to develop a rational and

Comparative Study of Data Compression Techniques

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ABSTRACT

This document discusses data compression and some of the data compression techniques. Data Compression is a technique of reducing the amount of space data occupies, to ease the process of storage and communication. This involves but is not limited to interpretation and elimination of redundancy in data. The fundamental process of compression involves using a well drafted technique to convert the actual data into the compressed data (smaller size). Depending upon how well a compression technique works and how much data can be regenerated from the compressed data given by a certain technique, the technique is classified as either as a lossy data compression technique or lossless data compression technique.

Keywords

Data Compression, Lossless Data Compression.

1. INTRODUCTION

Data compression is a process of converting a particular data into another form of representation which has a lesser storage requirement. Data compression is the process of applying some compression algorithm on a data or information for changing its bits representation in such way of it reduces the size of information while storing on disk. Data compression technique is used in the cloud computing mainly for the following two reasons:-

1. Increase in the transfer speed of data.
2. Decrease the storage space required to store data.

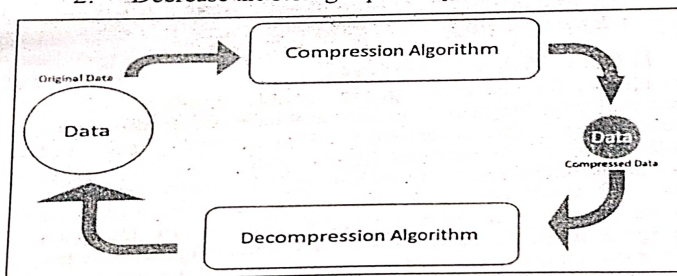


Figure 1: Pictorial Representation of Data Compression

In Lossless data compression, no actual information is lost because the bits are reduced by identifying the redundancy of data and eliminating only the redundancy. This technique is mostly applied for text document because any loss in a text document is acceptable. For example; zipping documents into one zip file compresses the data without any loss of a data while decompressing it. Lossless data compression technique is also known as a reversible compressing technique. Lossy

compression reduces data size by eliminating the less valuable or unessential part of the data i.e. In a lossy data compression technique when data is compressed, some data or information is permanently removed. As a result, while doing decompression on the compressed data, we get back data which does not match the original data and the loss of data is clearly visible to the user. This technique is used for the compressing some large media file such as audio, video, jpeg file. Mp3 is yet another form of compression technique applied on audio. For e.g.: while sharing images clicked on our smartphones with the high mega pixels (size of pictures are high) and sharing them on WhatsApp application, the application decompresses the image to increase the transfer speed between the sender and receiver, thus resulting in a lower quality mega pixel image at the receiver's end.

Example of Lossy Compression

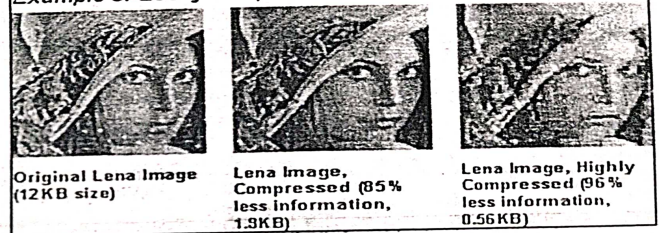


Figure 2: Levels of Lossy Compression

2. CLASSIFICATION OF LOSSLESS DATA COMPRESSION

Please Lossless data compression techniques can be classified further into Entropy based data compression technique and Dictionary based data compression technique.

In Entropy based data compression technique, the frequency of repeating data is first found and the repeating sequences are encoded.

In Dictionary based data compression, a dictionary is maintained and the encoder looks for any entry for every piece of data [4]. If there is an entry the encoder replaces the piece of data with the mapped encoding. If the entry is not present a new entry is made in the dictionary.

3. RUN-LENGTH ENCODING (RLE)

Please This is lossless data compression algorithm in which algorithm is applied on data in such way that large repeating pattern is caprice into small number of character of string. algorithm produce two bytes of output for each repeating character in data, one byte says that the total number character present in repeating pattern, and another bytes says that actual repeating character in string it's also called as run. (and its

Cambridge Analytica – A Case Study

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Abstract

Objectives: This study discusses how Facebook users' data has been harvested, used to formulate an algorithm to understand users' personality traits and in-turn use the process to influence the outcome of US Presidential Elections. **Method:** A Quiz application was developed to collect Users' data. Their activities on Social Media were analyzed, patterns were detected, OCEAN scores were given, and user groups were made on the basis of their political affiliation. And finally, they were targeted with suitable ads and news to achieve desirable results. **Findings/Application:** Use of Internet is increasing day-by-day and so is our Digital Footprints. Companies like Netflix started using these data to understand their Customers' behaviour and to improve their Customer Oriented Marketing Strategies. It has been observed that activities on Social Media say a lot about the users and sometimes it is used by companies harvesting user data in unethical ways.

Keywords: Cambridge Analytica, Facebook, OCEAN score, US Elections

1. Introduction

The use of information to manipulate people is an old concept referred to as 'theory of mind'^{1,2}. Though this concept has various perceived uses, an important use in today's socially connected world is to deceive. Deception is an evolutionary trait in some animals E.g. Camouflage or acquired gradually by learning in case of humans. This combined with the human intelligence makes the level of deception sophisticated or rather scary.

This study deals with a recent scandal involving Cambridge Analytica, Facebook and the US Elections, where known information was used to deceive and manipulate people in order to change their political views and votes.

Cambridge Analytica was found to be using Facebook Data sourced from a Cambridge University professor to work for a US Presidential Candidate. This malpractice was exposed by the former director of Cambridge Analytica, Christopher 'Chris' Wylie. The estimate of

user data breached ranges anywhere from 30 million ~ 80 million profiles.

2. Literature Survey

According to³ Cambridge University professor built a Facebook Quiz app that exploited a loophole in Facebook API that allowed collecting the App user's data as well as their friend's data. Although Facebook prohibited selling of user data collected via this method, it was sold and misused.

Medium in its case study has explained what and how psychology and behavior along with propaganda can manipulate people to do things intentionally or otherwise. They have linked it to the Cambridge Analytica Scandal by quoting how data driven marketing techniques can change behavior in target populations irrespective of the domain.

In⁴ study has defined what the data actually looked like and explained the process of how it worked from

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Optical Mark Recognition using Open CV

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ABSTRACT

The concept behind this paper is to develop a system that can check and evaluate the MCQ answers using the webcam. Till date many institution and organization held many exams where user are provided with a separate question and answer sheet, where answer are multiple choice option, each option may be a square or circle and user is supposed to either tick or fill it using pen or pencil. Such exams are JEE, CET, NEET etc. Also many financial institution and college refer such types of test since MCQ format Q&A are efficient and time saving as compared to that of lengthy papers. The renowned high level organization or institution use OMR technology for assessment of such paper, since number of student are in million and it's not feasible to check each single paper as it is time consuming, also it requires lot of human labor another problem is that human are prone to make error for example humans may grade wrong answer as correct / correct answer as wrong, Human may be biased about particular student and may score him less mark.

Maintaining paper for longer time is troublesome since it is prone to various climate change (fire, moisture etc) also humans have to manually record all the scores in different excel sheet/ file. On top of that high level organization use huge scanner which scans each paper in continuous batch precisely, where as normal school or college cannot afford such type of machine. Here the aim was to build a system such that when answer sheet is placed before webcam it will automatically get its image and all the answer will be evaluated and result will be displayed, determining score and total percentage. Since laptop is easily available to all, this make system more easy to handle and accessible to all as compared to other systems.

The problem statement is divided into 2 phases, First is how to extract image using 'open cv' and Second is how to apply content-filtration and image processing algorithm on it to get expected result.

Keywords

Optical Mark Recognition

1. INTRODUCTION

Conducting multiple choice questions examination worldwide has become one of the common means for evaluation of students credibility. Not only in India but also in other country various examinations are held which are in format of MCQ, In india JEE MAINS (Joint Entrance Exam) held by IIT, AIPMT (All Indian Pre-Medical Entrance Test) held by central board of ministry education, MHT-CET are such of the important exams that use this format to conduct examination. Also worldwide exams like TOEFL and GRE, GMAT are MCQ based. Reason for conducting MCQ exams is, it saves times in terms of time and evaluation, which makes it one of the efficient approach to conduct exams.

Every year millions of student appear for this test/ exam, where they read each question and they are supposed to mark or highlight there answer using pen or pencil. Currently high level organization that conduct exams like JEE, NEET have high configuration machine like special scanner which is costly, where this scanner are highly capable of capturing images and performs evaluation on further basis, but normal college cannot afford it. At present OMR technology is very expensive, not only because of the hardware requirement but also because of OMR software, as it's obvious that software that reduces the human effort and error paves the way to extra expense. Hence small school, institute cannot use this technology if they want to conduct normal test because of above challenges.

They resort to manually check the paper and note down all the answer of particular student and determine whether student has qualified or not. On an average based on number of question's evaluating each paper may take around 5 min. Hence if there are 100 students then it will take around 500 minutes or 8.3 hours of evaluation time which is not acceptable. The main purpose of our project was to save money as well as time and manpower, which is spent more by traditional approach, to overcome it. It's obvious that the OMR we are proposing does not match the efficiency and performance of high level OMR used by various institutes but it fulfill most of the need of various institution and organization and solves above problem. Now a days laptop are easily available and are one of the common mode of equipment used by all institutions, hence we thought that using laptop's webcam would be efficient approach to use for OMR. As webcam is builtin hardware and does not incur extra cost for installation of driver.

Here proposed OMR does following steps:

- ☐ Capture answer sheet image using webcam.
- ☐ Perform extraction of image page.
- ☐ Content filtration and apply image processing algorithm.
- ☐ Evaluate which answer is correct or incorrect and display total marks.
- ☐ After evaluation generate excel sheet and record all marks in it.

Below is image of answer sheet that will be used to evaluate

Agility of Supply Chain Management Solution Using Neural-Fuzzy Approach

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Enterprise solution is the architecture of collecting and processing business information. Business process agility affects process-based applications works as per changing business environment. This paper helps to understand different changing environment of business process in the supply chain domain. Changes depend on organizational policy; hence it can be incomplete or uncertain. To manage this unpredictable environment, a soft computing technique is used for constructing intelligent system. This paper shows use of Neuro-fuzzy approach to monitor agile behavior of the business process. Neural network phase is used for finding business process and parameter criticality. Fuzzy logic rule base phase calculates process agility based on the relation between process and its affecting parameter. Developed tool, shows that business architecture level is more prone to changes as compared to other architectural levels from the enterprise solution.

Keywords: Process Agility, Fuzzy Logic, Neural Network, SCM.

1. INTRODUCTION

Every organization has an enterprise solution which is differentiated on the basis of the architectural levels and process types. Supply chain management is the most agile system related to supplier, customer, product, and services it provides. In this paper, Section 2 show classification of enterprise solution as different types of processes with the different working architectural style. In Section 3, soft computing hybrid approach is defined to measure agility using Neuro-fuzzy approach. In Section 4, case study based on SCM domain is explained. This paper follows a practically oriented research methodology. Based on the actual study of enterprise architecture, organization processes are monitored for agility applicable at different architectural levels. The real-time case study is analyzed as per the architectural levels and process type specified in Section 1.

2. ENTERPRISE ARCHITECTURE

Enterprise architecture (EA) is a well-defined and well-organized practice of conducting analysis, design, and implementation of the software solution in conducting successful execution of business strategy [7, 10]. EA gives a formal explanation of applications at the different

component level, and guideline for governance. EA can be classified into the architectural level showing communication of all architectural levels in the enterprise solution. The business architecture uses requirement of information to implement application architecture. Information architecture works or changes its style based on application architecture. Data is provided from data architecture. Using technology architecture, data is stored and processed for application architecture and data architecture.

2.1. Business Processes Used in Enterprise Solution

Enterprise solution is the flow business processes as per the requirement [18, 20]. It may have different business collaboration structure, pattern and its behaviors.

A business process is designed as per the business rules w.r.t its objectives, pre- and post-business conditions [6, 24]. Figure 1 show the relationship of management, core, enabling and enhancing type of processes in the enterprise solutions. Management processes can be defined at all architectural levels where ever business logic is needed [15]. Core process defined at all enterprise levels for developing new or for maintaining existing business logic. Enabling and enhancing processes are for performance enhancement in sequence with core and management processes. Agility is not fixed or specific. It has metamorphic abilities which can be modified as per change in the business environment [14]. Agility is the capability

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Performance-based Seismic Evaluation of Different Concrete Bridges

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Abstract

In this study, analytical fragility curves are developed for the sample concrete bridges of Indian highways to be used in the assessment of their seismic vulnerability. Bridges are first grouped into certain bridge classes based on their structural attributes and sample bridges are generated to account for the structural variability. Nonlinear static analyses and incremental dynamic analyses are conducted for each bridge sample with their detailed 3-D analytical models under different earthquake ground motions having varying seismic intensities. Engineering demand parameters are employed in the determination of seismic response of the bridge components as well as defining damage limit states in terms of member capacities. Fragility curves are obtained from the probability of exceeding each specified damage limit state for each bridge class. Thus, for concrete bridges evaluation of seismic damage is carried out which is most important part to decide seismic vulnerability. Concrete bridges designed as per current Indian seismic codes with shear failure criteria based on capacity design concept are found to be the most vulnerable ones using performance base earthquake engineering. This research highlights urgent need of incorporating capacity design concept based on flexural failure criteria and performance base earthquake engineering in Indian seismic codes. Developed fragility curves can be implemented in the seismic risk assessment packages for mitigation purposes.

Keywords: concrete bridge, seismic, pushover, fragility

1. INTRODUCTION

Bridges are essential elements in modern transportation network and play a significant role in a country's economy. However, it has always been a major challenge to keep bridges safe and serviceable. Modern bridge design codes include seismic detailing in order to ensure ductile behavior. The main parameters affecting the performance of bridge (tie spacing, concrete and steel properties, amount of reinforcement) varies significantly from old to modern bridges. Seismic vulnerability assessment of the highway bridges located within earthquake prone regions and determination of their performance levels under seismic actions play an important role for the safety of transportation systems [1].

The goal of this study is to refine an existing methodology for the development of analytical seismic fragility curves for bridge classes that are typical to the Indian region. India is a moderate seismic zone in which the seismic risk mitigation effort has lagged the effort made by the Western countries.

More focus is now being placed upon the Zone-V of India because of the recognition of potential for large ground shaking. To aid in the assessment of seismic risk for the region, typical bridge classes have been selected for the development of seismic fragility curves. The level of detail that is used and the refinement of the analytical methodology will aid in generating more reliable fragility curves than what are currently available. The purpose of this study is to use analytical methods to generate fragility curves for different bridge classes which are most common to the Indian perspective.

It has been observed that under the recent earthquake excitation, many bridges were either totally collapsed or received major damages and demonstrated the severe vulnerability. Even a moderate damage to such structures will make them non-functional. It may seriously affect the major relief work after earthquake. A different approach for protection of bridges from earthquakes is evaluating seismic performance and suggesting retrofit techniques using latest proven methods of performance analysis [2]. The concept of seismic performance analysis of new bridges and the assessment and retrofit of existing bridges was practically used in America, New Zealand, Canada, China, England, France, Japan, Italy and Mexico etc. The research on evaluation of seismic damage to concrete bridges has great importance, to decide correct approach and methodology for performance base design and it has great concerned in India. Therefore, evaluation of seismic damage to concrete bridges designed as per current Indian Standards is carried out in the present study. The bridge is first analysed using nonlinear static analysis (pushover analysis) by SAP2000 [3]. The seismic performance and overall seismic capacity of the bridge are then investigated through incremental dynamic analyses (IDA) by developing IDA curves [4] using IDARC2D software [5]. Nonlinear dynamic analysis is applied to simulate the earthquake loads over the analytical model. A set of ten Indian ground motions is selected. Fragility curves are generated to define the damage limit states and probability of survival of the structure.